Drag Drop Questions

14 questions | Original English questions and source answers

Question 1

Refer to the exhibit.

```
[root#HostTime=]#ip route
default via 192.168.1.193 dev eth1 proto static
192.168.1.0/26 dev sth1 proto kernel scope link src 192.168.1.200 metric 1

[root#HostTime=]#ip addr show eth1
eth1:mtu 1500 qdisc pfifo_fast qlan 1000
 link/ether 00:0C:22:83:79:A3 brd ff:ff:ff:ff:ff:ff
 inet 192.168.1.200/26 hrd 192.168.1.255 scope global eth1
 inet6 fe80::20c::29ff:fe89:79b3/64 scope link
 valid_lft forever preferred_lft forever
```

Drag and drop the networking parameters from the left onto the correct values on the right.

Available items

NIC vendor OUI

NIC MAC address

default gateway

host IP address

subnet mask

Descriptions / target positions

00:0C:22

00:0C:22:83:79:A3

192.168.1.193

192.168.1.200

255.255.255.192

Answer

Description / position	Source answer
00:0C:22	NIC vendor OUI
00:0C:22:83:79:A3	NIC MAC address
192.168.1.193	default gateway
192.168.1.200	host IP address
255.255.255.192	subnet mask

Source explanation

+ 00:0C:22: NIC vendor OUI
+ 00:0C:22:83:79:A3: NIC MAC address
+ 192.168.1.193: default gateway
+ 192.168.1.200: host IP address
+ 255.255.255.192: subnet mask

The “ip route” and “ip addr show eth1” are Linux commands.
+ “ip route”: display the routing table
+ “ip addr show eth1”: get depth information (only on eth1 interface) about your network interfaces like IP Address, MAC Address information

Source: CCNA 200-301 Questions.html | Question 1

Question 2

Drag and drop the functions from the left onto the correct network components on the right.

Available items

DHCP Server

DNS Server

Descriptions / target positions

1. holds the TCP/IP settings to be distributed to the clients

2. resolves web URLs to IP addresses

3. stores a list of IP addresses mapped to names

4. assigns a default gateway to a client

5. assigns IP addresses to enabled clients

Answer

Description / position	Source answer
1. holds the TCP/IP settings to be distributed to the clients	DHCP Server
2. resolves web URLs to IP addresses	DNS Server
3. stores a list of IP addresses mapped to names	DNS Server
4. assigns a default gateway to a client	DHCP Server
5. assigns IP addresses to enabled clients	DHCP Server

Source: CCNA 200-301 Questions.html | Question 2

Question 3

Drag the OSPF parameters to the correct places on the right. Note: There is one unused answer in this question.

Available items

must be unique

must match

unused answer

Descriptions / target positions

1. netmask

2. OSPF process ID

3. router ID

4. IP address

5. area ID

6. timers

Answer

Description / position	Source answer
1. netmask	must match
2. OSPF process ID	unused answer
3. router ID	must be unique
4. IP address	must be unique
5. area ID	must match
6. timers	must match

Source: CCNA 200-301 Questions.html | Question 3

Question 4

Drag and drop the IPv4 network subnets from the left onto the correct usable host ranges on the right.

Available items

172.28.192.1 – 172.28.255.254

172.28.224.1 – 172.28.231.254

172.28.228.1 – 172.28.229.254

172.28.228.129 – 172.28.228.254

172.28.228.145 – 172.28.228.150

Descriptions / target positions

172.28.228.144/18

172.28.228.144/21

172.28.228.144/23

172.28.228.144/25

172.28.228.144/29

Answer

Description / position	Source answer
172.28.228.144/18	172.28.192.1 – 172.28.255.254
172.28.228.144/21	172.28.224.1 – 172.28.231.254
172.28.228.144/23	172.28.228.1 – 172.28.229.254
172.28.228.144/25	172.28.228.129 – 172.28.228.254
172.28.228.144/29	172.28.228.145 – 172.28.228.150

Source explanation

+ 172.28.228.144/18: 172.28.192.1 – 172.28.255.254
+ 172.28.228.144/21: 172.28.224.1 – 172.28.231.254
+ 172.28.228.144/23: 172.28.228.1 – 172.28.229.254
+ 172.28.228.144/25: 172.28.228.129 – 172.28.228.254
+ 172.28.228.144/29: 172.28.228.145 – 172.28.228.150

This subnet question requires us to grasp how to subnet very well. To quickly find out the subnet range, we have to find out the increment and the network address of each subnet. Let’s take an example with the subnet 172.28.228.144/18:

From the /18 (= 1100 0000 in the 3rd octet), we find out the increment is 64. Therefore the network address of this subnet must be the greatest multiple of the increment but not greater than the value in the 3rd octet (228). We can find out the 3rd octet of the network address is 192 (because 192 = 64 * 3 and 192 < 228) -> The network address is 172.28.192.0. So the first usable host should be 172.28.192.1 and it matches with the 5th answer on the right. In this case we don’t need to calculate the broadcast address because we found the correct answer.

Let’s take another example with subnet 172.28.228.144/23 -> The increment is 2 (as /23 = 1111 1110 in 3rd octet). The 3rd octet of the network address is 228 (because 228 is the multiply of 2 and equal to the 3rd octet) -> The network address is 172.28.228.0 -> The first usable host is 172.28.228.1. It is not necessary but if we want to find out the broadcast address of this subnet, we can find out the next network address, which is 172.28.(228 + the increment number).0 or 172.28.230.0 then reduce 1 bit -> 172.28.229.255 is the broadcast address of our subnet. Therefore the last usable host is 172.28.229.254.

If you are still unclear about how to do subnetting quickly, please read part 3 of our “Subnetting Tutorial – Subnetting Made Easy” at https://www.9tut.com/subnetting-tutorial/3

Source: CCNA 200-301 Questions.html | Question 4

Question 5

Drag and drop the threat-mitigation techniques from the left onto the types of threat or attack they mitigate on the right.

Available items

Configure a VLAN access control list

Configure the dynamic ARP inspection feature

Configure the root guard feature

Configure the BPDU guard feature

Descriptions / target positions

802.1q double tagging

ARP spoofing

unwanted superior BPDUs

unwanted BPDUs on PortFast-enabled interfaces

Answer

Description / position	Source answer
802.1q double tagging	Configure a VLAN access control list
ARP spoofing	Configure the dynamic ARP inspection feature
unwanted superior BPDUs	Configure the root guard feature
unwanted BPDUs on PortFast-enabled interfaces	Configure the BPDU guard feature

Source explanation

+ 802.1q double tagging: Configure VACL
+ ARP spoofing: Configure dynamic ARP inspection
+ unwanted superior BPDUs: Configure root guard
+ unwanted BPDUs on PortFast-enabled interfaces: Configure BPDU guard

Double-Tagging attack:

Attacker --VLAN 10--> Company Switch A --Native VLAN 10--> Company Switch B --VLAN 20--> Victim [Payload|VLAN 20|VLAN 10] [Payload|VLAN 20] [Payload]

In this attack, the attacking computer generates frames with two 802.1Q tags. The first tag matches the native VLAN of the trunk port (VLAN 10 in this case), and the second matches the VLAN of a host it wants to attack (VLAN 20).

When the packet from the attacker reaches Switch A, Switch A only sees the first VLAN 10 and it matches with its native VLAN 10 so this VLAN tag is removed. Switch A forwards the frame out all links with the same native VLAN 10. Switch B receives the frame with an tag of VLAN 20 so it removes this tag and forwards out to the Victim computer.

Note: This attack only works if the trunk (between two switches) has the same native VLAN as the attacker.

To mitigate this type of attack, you can use VLAN access control lists (VACLs, which applies to all traffic within a VLAN. We can use VACL to drop attacker traffic to specific victims/servers) or implement Private VLANs.

ARP attack (like ARP poisoning/spoofing) is a type of attack in which a malicious actor sends falsified ARP messages over a local area network as ARP allows a gratuitous reply from a host even if an ARP request was not received. This results in the linking of an attacker’s MAC address with the IP address of a legitimate computer or server on the network. This is an attack based on ARP which is at Layer 2. Dynamic ARP inspection (DAI) is a security feature that validates ARP packets in a network which can be used to mitigate this type of attack.

Source: CCNA 200-301 Questions.html | Question 5

Question 6

Drag and drop the description of file-transfer protocols from the left onto the correct protocols on the right.

Available items

FTP

TFTP

Descriptions / target positions

1. does not require user authentication

2. uses UDP

3. uses TCP

4. uses ports 20 and 21

5. provides reliability when loading an IOS image upon boot up

6. uses port 69

Answer

Description / position	Source answer
1. does not require user authentication	TFTP
2. uses UDP	TFTP
3. uses TCP	FTP
4. uses ports 20 and 21	FTP
5. provides reliability when loading an IOS image upon boot up	FTP
6. uses port 69	TFTP

Source: CCNA 200-301 Questions.html | Question 6

Question 7

Drag and drop the application protocols from the left onto the suitable transport protocols on the right.

Available items

TCP

UDP

Descriptions / target positions

1. SMTP
2. SNMP
3. DHCP
4. TFTP
5. FTP
6. SSH

Answer

Description / position	Source answer
1. SMTP	TCP
2. SNMP	UDP
3. DHCP	UDP
4. TFTP	UDP
5. FTP	TCP
6. SSH	TCP

Source: CCNA 200-301 Questions.html | Question 7

Question 8

Drag and drop the descriptions of IP protocol transmissions from the left onto the correct IP traffic types on the right.

Available items

TCP

UDP

Descriptions / target positions

1. Sends Transmission in Sequence

2. Transmits packets individually

3. Uses higher transmission rate to support latency-sensitive apps

4. Uses a lower transmission rate to ensure reliability

5. Transmissions include an 8-byte header

6. Transmits packet as stream

Answer

Description / position	Source answer
1. Sends Transmission in Sequence	TCP
2. Transmits packets individually	UDP
3. Uses higher transmission rate to support latency-sensitive apps	UDP
4. Uses a lower transmission rate to ensure reliability	TCP
5. Transmissions include an 8-byte header	UDP
6. Transmits packet as stream	TCP

Source explanation

UDP is a simpler message-based connectionless protocol. In connectionless protocols, there is no effort made to setup a dedicated end-to-end connection.

...

Datagrams - Packets are sent individually and are guaranteed to be whole if they arrive. Packets have definite bounds and no split or merge into data streams may exist.

According to https://community.cisco.com/t5/networking-documents/udp/ta-p/3114870

TCP is connection-oriented and UDP is connectionless. This means that before sending TCP packets, a connection is established between the server and the client. This process of setting up a connection is called TCP handshaking. The stream of packets is then sent over this connection -> TCP transmits packet as stream.

Reference: https://www.vpnmentor.com/blog/tcp-vs-udp/

Source: CCNA 200-301 Questions.html | Question 8

Question 9

Drag and drop the WLAN components from the left onto the correct descriptions on the right.

Available items

wireless LAN controller

access point

service port

virtual interface

dynamic interface

Descriptions / target positions

device that manages access points

device that provides Wi-Fi devices with a connection to a wired network

used for out of band management of a WLC

used to support mobility management of the WLC

applied to the WLAN for wireless client communication

Answer

Description / position	Source answer
device that manages access points	wireless LAN controller
device that provides Wi-Fi devices with a connection to a wired network	access point
used for out of band management of a WLC	service port
used to support mobility management of the WLC	virtual interface
applied to the WLAN for wireless client communication	dynamic interface

Source explanation

+ device that manages access points: wireless LAN controller
+ device that provides Wi-Fi devices with a connection to a wired network: access point
+ used for out of band management of a WLC: service port
+ used to support mobility management of the WLC: virtual interface
+ applied to the WLAN for wireless client communication: dynamic interface

The service port can be used management purposes, primarily for out-of-band management. However, AP management traffic is not possible across the service port. In most cases, the service port is used as a “last resort” means of accessing the controller GUI for management purposes. For example, in the case where the system distribution ports on the controller are down or their communication to the wired network is otherwise degraded.

A dynamic interface with the Dynamic AP Management option enabled is used as the tunnel source for packets from the controller to the access point and as the destination for CAPWAP packets from the access point to the controller.

The virtual interface is used to support mobility management, Dynamic Host Configuration Protocol (DHCP) relay, and embedded Layer 3 security such as guest web authentication. It also maintains the DNS gateway host name used by Layer 3 security and mobility managers to verify the source of certificates when Layer 3 web authorization is enabled.

Reference:
https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-5/config-guide/b_cg85/ports_and_interfaces.html

Source: CCNA 200-301 Questions.html | Question 9

Question 10

Drag and drop the networking parameters from the left on to the correct values on the right.

Available items

Connection Oriented

Connectionless

Descriptions / target positions

1. SMTP
2. SNMP
3. TFTP
4. VoIP
5. SSH
6. FTP

Answer

Description / position	Source answer
1. SMTP	Connection Oriented
2. SNMP	Connectionless
3. TFTP	Connectionless
4. VoIP	Connectionless
5. SSH	Connection Oriented
6. FTP	Connection Oriented

Source explanation

SSH uses TCP port 22 while SNMP uses UDP port 161 and 162.

Source: CCNA 200-301 Questions.html | Question 10

Question 11

Drag drop the descriptions from the left on to the correct configuration-management technologies on the right.

Available items

Ansible

Chef

Puppet

Descriptions / target positions

1. uses Ruby for fundamental configuration elements

2. uses TCP port 10002 for configuration push jobs

3. uses SSH for remote device communication

4. fundamental configuration elements are stored in a manifest

5. uses TCP 8140 for communication

6. uses YAML for fundamental configuration elements

Answer

Description / position	Source answer
1. uses Ruby for fundamental configuration elements	Chef
2. uses TCP port 10002 for configuration push jobs	Chef
3. uses SSH for remote device communication	Ansible
4. fundamental configuration elements are stored in a manifest	Puppet
5. uses TCP 8140 for communication	Puppet
6. uses YAML for fundamental configuration elements	Ansible

Source explanation

The focus of Ansible is to be streamlined and fast, and to require no node agent installation. Thus, Ansible performs all functions over SSH. Ansible is built on Python, in contrast to the Ruby foundation of Puppet and Chef.

TCP port 10002 is the command port. It may be configured in the Chef Push Jobs configuration file . This port allows Chef Push Jobs clients to communicate with the Chef Push Jobs server.

Puppet is an open-source configuration management solution, which is built with Ruby and offers custom Domain Specific Language (DSL) and Embedded Ruby (ERB) templates to create custom Puppet language files, offering a declarative-paradigm programming approach.

A Puppet piece of code is called a manifest, and is a file with .pp extension.

Source: CCNA 200-301 Questions.html | Question 11

Question 12

Drag and drop the AAA functions from the left onto the correct AAA services on the right.

Available items

Authentication

Authorization

Accounting

Descriptions / target positions

1. records user activities
2. restricts the services that are available to a user
3. identifies the user
4. controls the actions that a user can perform
5. provides analytical information for the network administrator
6. verifies the password associated with a user

Answer

Description / position	Source answer
1. records user activities	Accounting
2. restricts the services that are available to a user	Authorization
3. identifies the user	Authentication
4. controls the actions that a user can perform	Authorization
5. provides analytical information for the network administrator	Accounting
6. verifies the password associated with a user	Authentication

Source: CCNA 200-301 Questions.html | Question 12

Question 13

Drag and Drop the benefits of a Cisco Wireless LAN Controller from the left onto the correct examples on the right.

Available items

Access points auto adjust signal strength

Controller provides centralized management of users and VLANs

Controller uses loadbalancing to maximize throughput

Controller image auto deployed to access Points

Descriptions / target positions

Dynamic RF Feature

Easy Deployment Process

Optimized user performance

Easy upgrade process

Answer

Description / position	Source answer
Dynamic RF Feature	Access points auto adjust signal strength
Easy Deployment Process	Controller provides centralized management of users and VLANs
Optimized user performance	Controller uses loadbalancing to maximize throughput
Easy upgrade process	Controller image auto deployed to access Points

Source explanation

+ Dynamic RF Feature: Access points auto adjust signal strength
+ Easy Deployment Process: Controller provides centralized management of users and VLANs
+ Optimized user performance: Controller uses loadbalancing to maximize throughput
+ Easy upgrade process: Controller image auto deployed to access Points

Source: CCNA 200-301 Questions.html | Question 13

Question 14

Drag and drop the Cisco Wireless LAN Controller security settings from the left onto the correct security mechanism categories on the right.

Available items

Layer 2 Security Mechanism

Layer 3 Security Mechanisms (for WLAN)

Descriptions / target positions

1. web policy

2. Passthrough

3. WPA+WPA2

4. 802.1X

Answer

Description / position	Source answer
1. web policy	Layer 3 Security Mechanisms (for WLAN)
2. Passthrough	Layer 3 Security Mechanisms (for WLAN)
3. WPA+WPA2	Layer 2 Security Mechanism
4. 802.1X	Layer 2 Security Mechanism

Source explanation

Layer 2 Security Mechanism includes WPA+WPA2, 802.1X, Static WEP, CKIP while Layer 3 Security Mechanisms (for WLAN) includes IPSec, VPN Pass-Through, Web Passthrough …

Reference: https://www.cisco.com/c/en/us/support/docs/wireless/4400-series-wireless-lan-controllers/106082-wlc-compatibility-matrix.html

Source: CCNA 200-301 Questions.html | Question 14